

Linear Programming Report

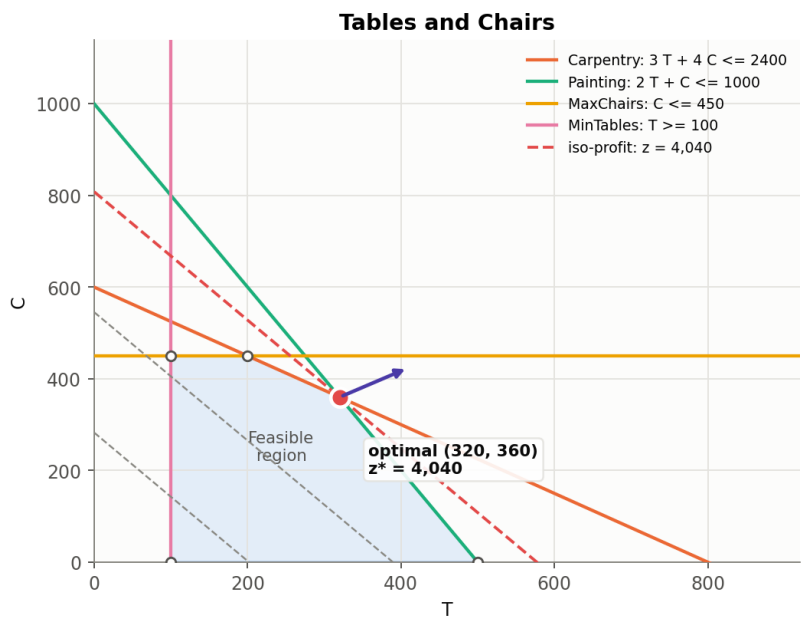
Sensitivity Analysis & Duality

Tables and Chairs

A shop produces tables (T) and chairs (C). Maximize profit subject to carpentry and painting hours, a market limit on chairs, and a minimum number of tables.

Source: Lecture slides 6-19 (graphical sensitivity analysis)

Problem type	MAX, 2 variables, 4 constraints
Status	OPTIMAL
Optimal Profit (\$)	4,040
Dual status	OPTIMAL
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Course	Sensitivity Analysis and Duality (Dr. Islam Ali)
Date	27 September 2026, 15:11



1. Model formulation

MAX $z = 7T + 5C$
 subject to
 $3T + 4C \leq 2400$ (Carpentry)
 $2T + C \leq 1000$ (Painting)
 $C \leq 450$ (MaxChairs)
 $T \geq 100$ (MinTables)
 $T \geq 0, C \geq 0$

Data table

Constraint	T	C	Sense	RHS
Carpentry	3	4	$\leq$	2,400
Painting	2	1	$\leq$	1,000
MaxChairs	0	1	$\leq$	450
MinTables	1	0	$\geq$	100
Objective c	7	5	max	—

Definitions

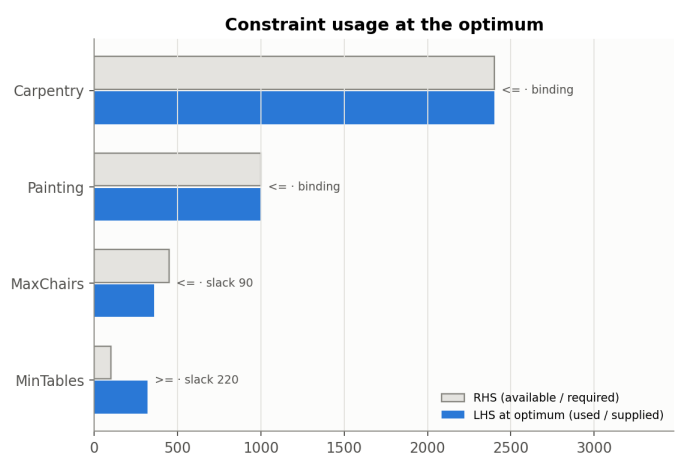
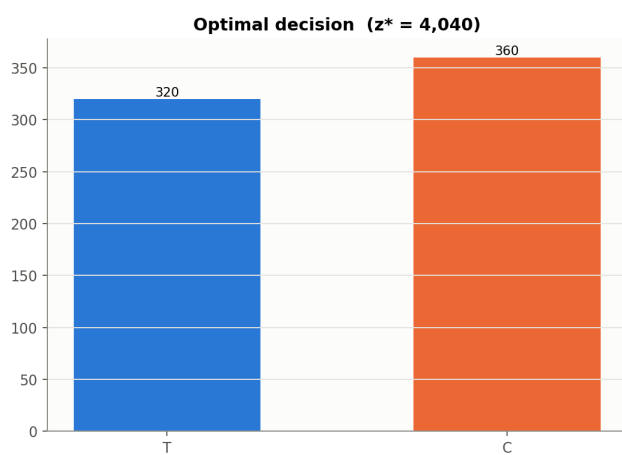
- **T**: number of tables produced
- **C**: number of chairs produced
- **Carpentry**: carpentry hours available
- **Painting**: painting hours available
- **MaxChairs**: maximum number of chairs
- **MinTables**: minimum number of tables

2. Optimal solution

Optimal Profit (\$): **4,040**

Variable	Description	Optimal value	Obj. coefficient	Contribution
T	number of tables produced	320	7	2,240
C	number of chairs produced	360	5	1,800

Constraint	LHS	Sense	RHS	Slack/Surplus	Binding
Carpentry	2,400	<=	2,400	0	yes
Painting	1,000	<=	1,000	0	yes
MaxChairs	360	<=	450	90	no
MinTables	320	>=	100	220	no



Corner points

T	C	Profit (\$)	Optimal?
100	0	700	
500	0	3,500	
320	360	4,040	✓
200	450	3,650	
100	450	2,950	

3. Sensitivity analysis

Objective-coefficient ranging (variable cells)

Variable	Value	Reduced cost	Obj. coef	Allow. +	Allow. –	Low	High
T	320	0	7	3	3.25	3.75	10
C	360	0	5	4.333	1.5	3.5	9.333

Right-hand-side ranging (constraints)

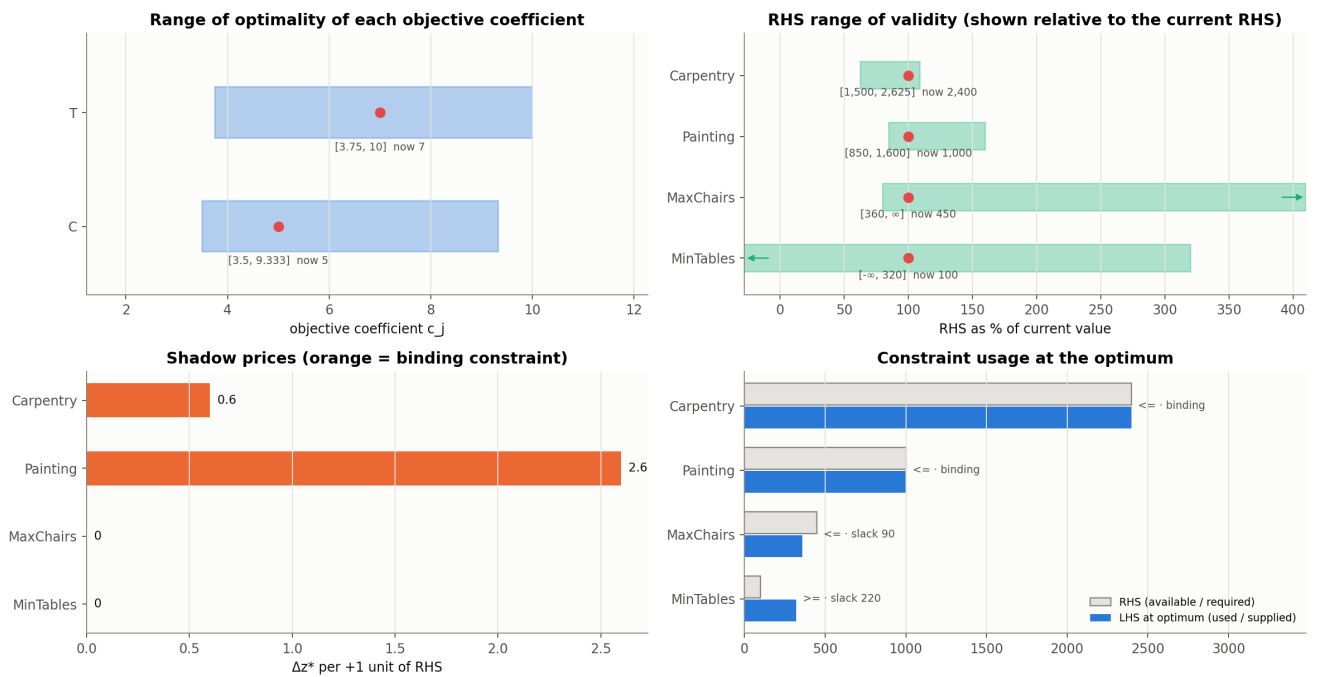
Constraint	LHS	Sense	RHS	Slack	Shadow price	Allow. +	Allow. –	Low	High
Carpentry	2,400	<=	2,400	0	0.6	225	900	1,500	2,625
Painting	1,000	<=	1,000	0	2.6	600	150	850	1,600
MaxChairs	360	<=	450	90	0	∞	90	360	∞
MinTables	320	>=	100	220	0	220	∞	$-\infty$	320

Interpretation

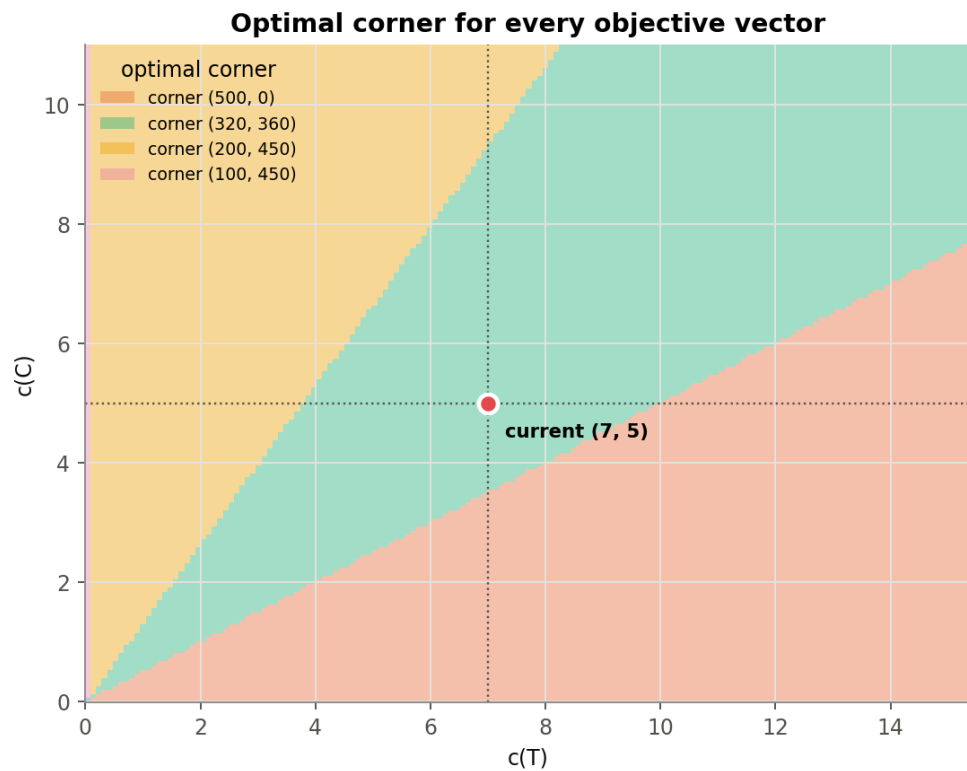
- **T = 320.** Its objective coefficient (7) may vary in [3.75, 10] without changing the optimal solution (one change at a time).
- **C = 360.** Its objective coefficient (5) may vary in [3.5, 9.333] without changing the optimal solution (one change at a time).
- **Carpentry** is binding. Shadow price 0.6: each extra unit of RHS changes Profit (\$) by 0.6, valid while the RHS stays in [1,500, 2,625].
- **Painting** is binding. Shadow price 2.6: each extra unit of RHS changes Profit (\$) by 2.6, valid while the RHS stays in [850, 1,600].
- **MaxChairs** is not binding (slack 90), so its shadow price is 0; the RHS can move within [360, ∞] without effect on Profit (\$).
- **MinTables** is not binding (slack 220), so its shadow price is 0; the RHS can move within [$-\infty$, 320] without effect on Profit (\$).

Sensitivity dashboard

Tables and Chairs: sensitivity dashboard ($z^* = 4,040$)

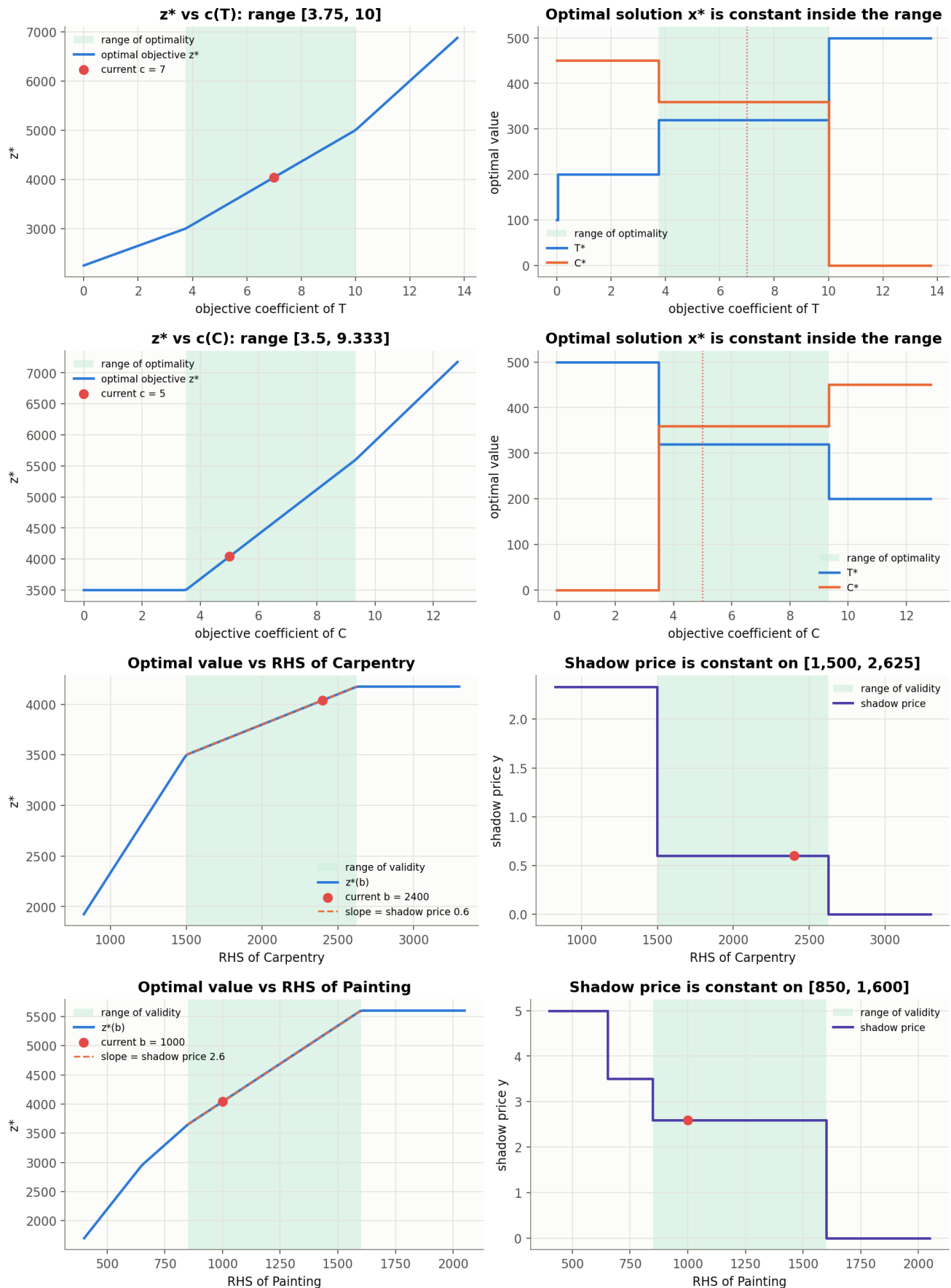


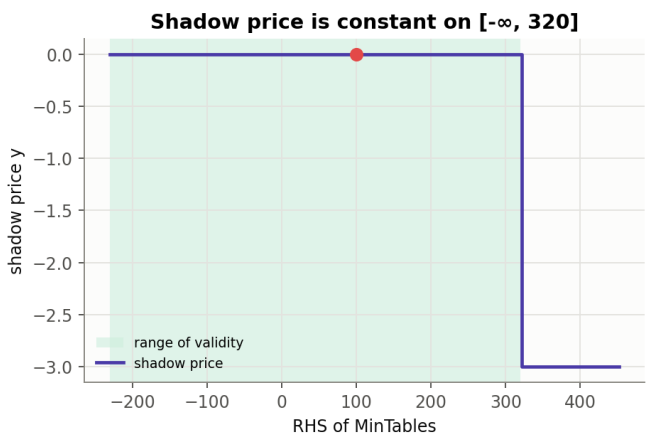
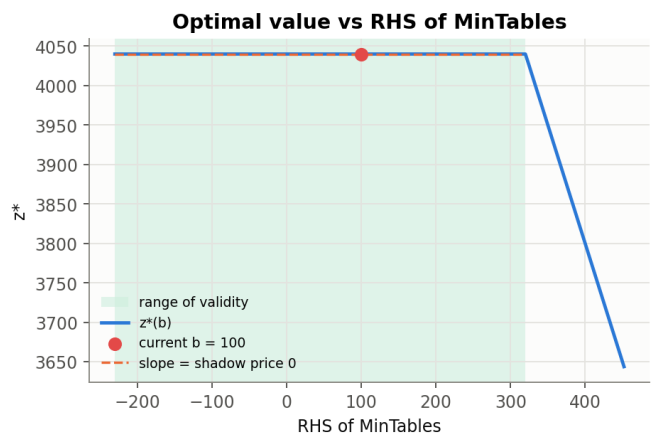
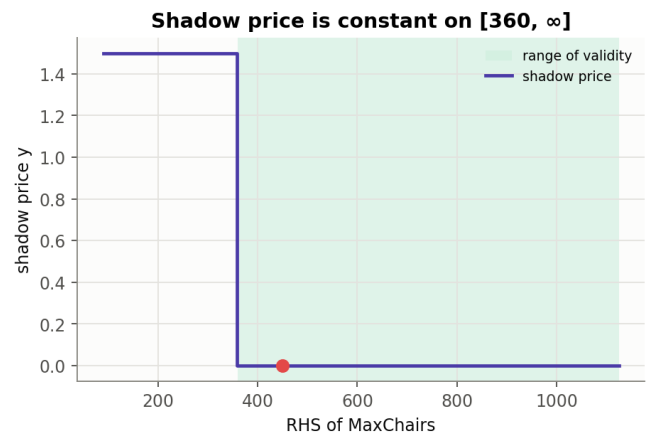
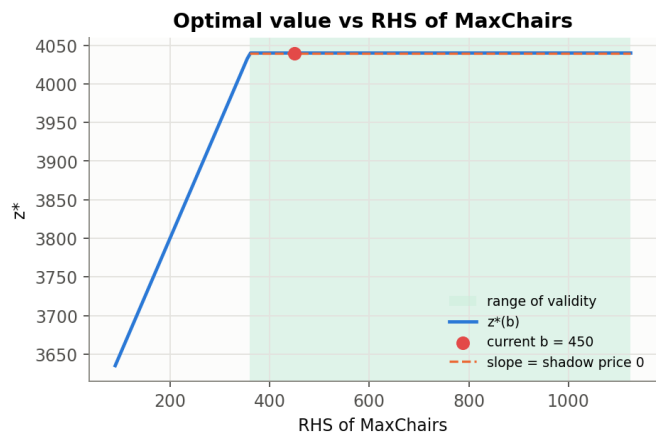
Optimal corner for every objective vector



4. Parametric analysis

Each chart re-solves the LP for a range of values of one coefficient. The shaded band is the range reported in section 3.



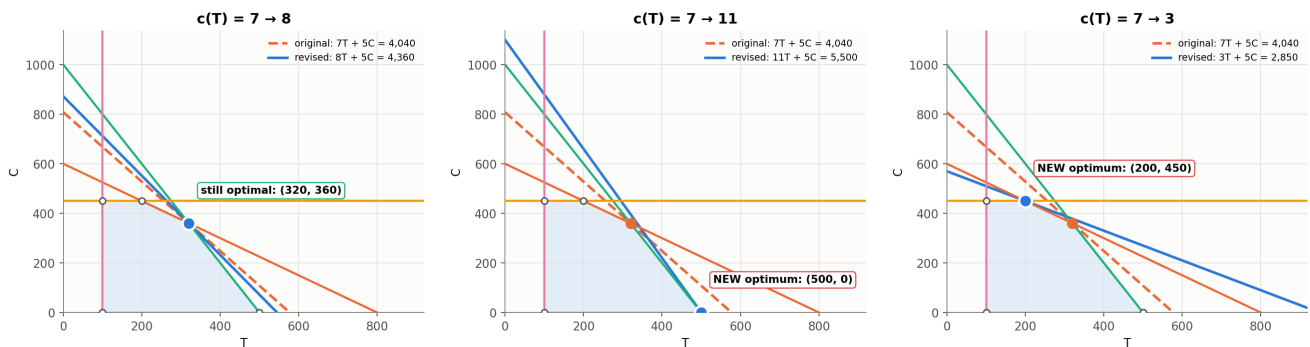


5. What-if scenarios

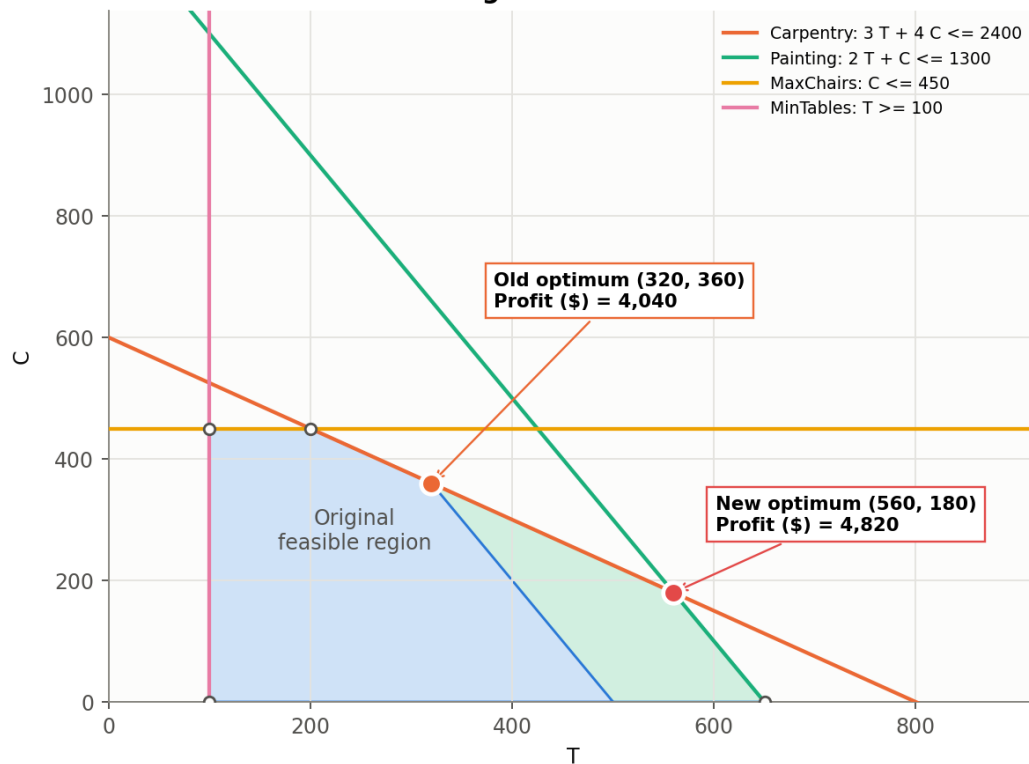
'Predicted' uses only the sensitivity report (shown when the change is inside its range); 'Actual' re-solves the LP.

Scenario	InRange	Predicted	Actual	NewSolution	Note
c(T): 7 → 8	yes	4,360	4,360	320, 360	slide 8/10: profit per table 7 → 8
c(T): 7 → 11	no	–	5,500	500, 0	slide 11: profit per table 11
c(T): 7 → 3	no	–	2,850	200, 450	slide 11: profit per table 3
b(Painting): 1000 → 1300	yes	4,820	4,820	560, 180	slide 14/16: painting hours 1000 → 1300

Changing an objective coefficient rotates the iso-profit line; the feasible region never moves



Painting: RHS 1000 → 1300



6. Duality

Dual formulation

MIN $z = 2400 y_{\text{Carpentry}} + 1000 y_{\text{Painting}} + 450 y_{\text{MaxChairs}} + 100 y_{\text{MinTables}}$
 subject to
 $3 y_{\text{Carpentry}} + 2 y_{\text{Painting}} + y_{\text{MinTables}} \geq 7 \quad (\text{T})$
 $4 y_{\text{Carpentry}} + y_{\text{Painting}} + y_{\text{MaxChairs}} \geq 5 \quad (\text{C})$
 $y_{\text{Carpentry}} \geq 0, y_{\text{Painting}} \geq 0, y_{\text{MaxChairs}} \geq 0, y_{\text{MinTables}} \leq 0$

Dual solution (resource / requirement prices)

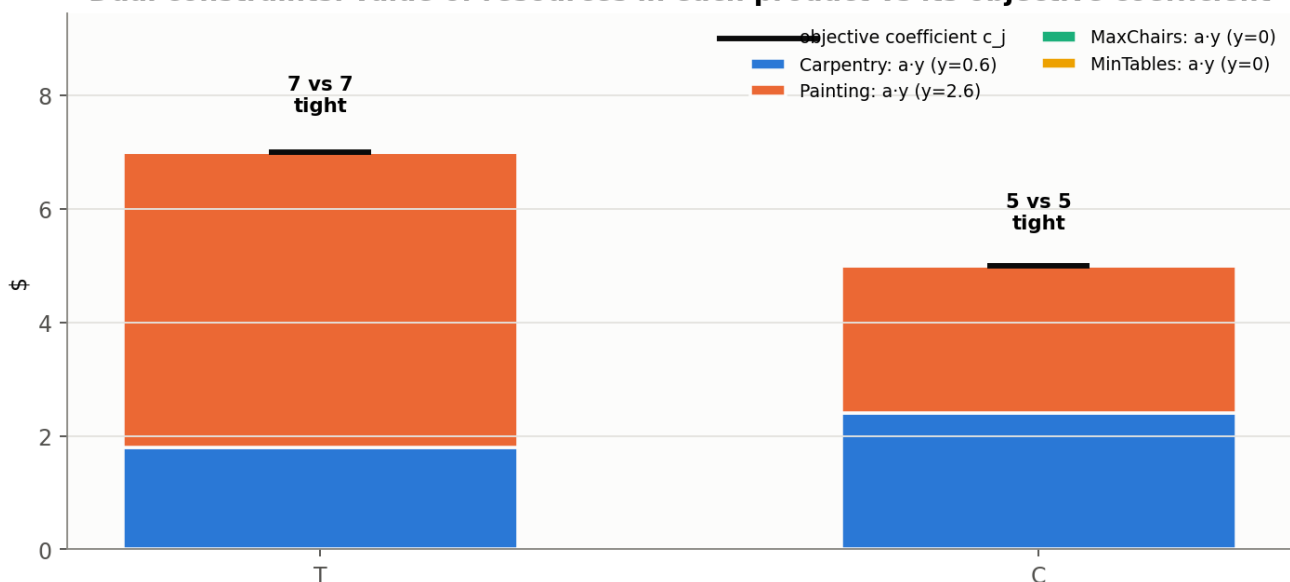
Dual variable	Primal constraint	Value y^*	Primal shadow price	$b_i \cdot y_i$
$y_{\text{Carpentry}}$	Carpentry	0.6	0.6	1,440
y_{Painting}	Painting	2.6	2.6	2,600
$y_{\text{MaxChairs}}$	MaxChairs	0	0	0
$y_{\text{MinTables}}$	MinTables	0	0	0

Strong duality check: primal $z^* = 4,040$, dual $w^* = 4,040$ (equal ✓). Dual of the dual equals the primal: yes.

Complementary slackness (all products must be 0)

Pair	Slack	DualValue	Product
Carpentry slack $\times y_{\text{Carpentry}}$	0	0.6	0
Painting slack $\times y_{\text{Painting}}$	0	2.6	0
MaxChairs slack $\times y_{\text{MaxChairs}}$	90	0	0
MinTables slack $\times y_{\text{MinTables}}$	220	0	0
$T \times \text{surplus of dual constraint T}$	0	320	0
$C \times \text{surplus of dual constraint C}$	0	360	0

Dual constraints: value of resources in each product vs its objective coefficient

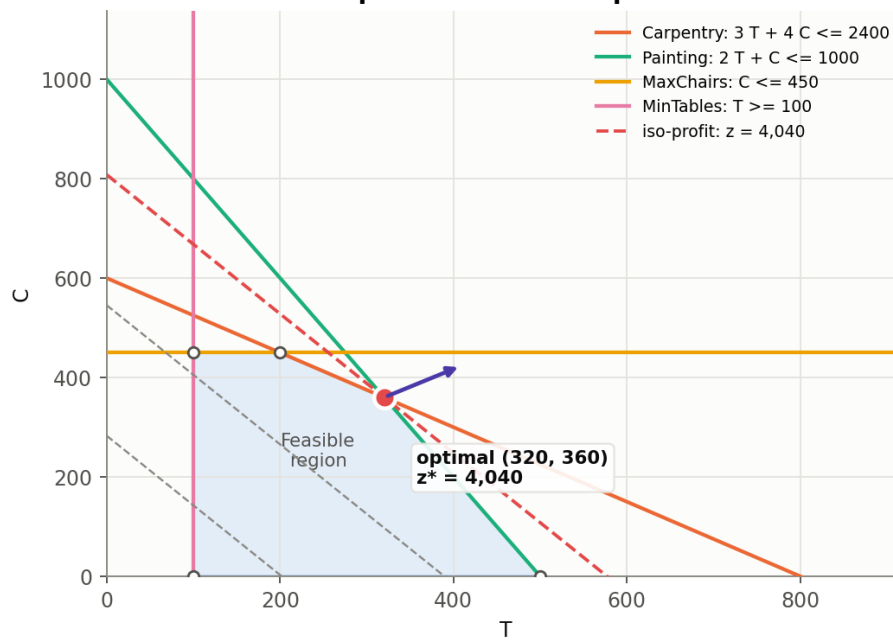


Primal (max): 4 constraints × 2 variables

7	5	← objective
3	4	≤ 2400
2	1	≤ 1000
0	1	≤ 450
1	0	≥ 100
T	C	

Dual (min): 2 constraints × 4 variables

2400	1000	450	100	← objective
3	2	0	1	≥ 7
4	1	1	0	≥ 5
y_Carpentry	y_Painting	y_MaxChairs	y_MinTables	

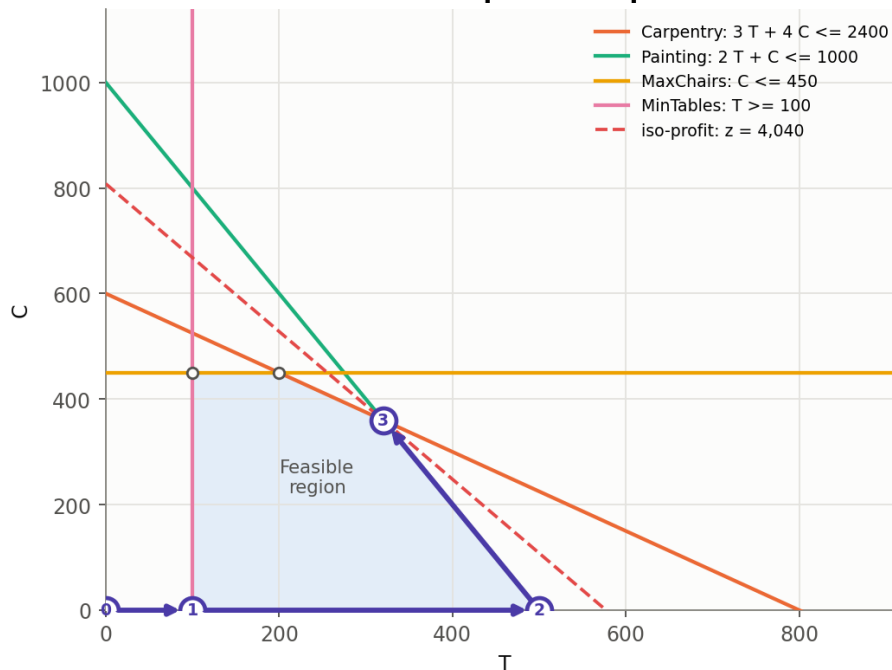
Graphical view of the primal

7. Simplex iterations

Two-phase tableau simplex (entering column = most negative $z_i - c_i$; leaving row = minimum ratio).

Tableau	Method	Phase	Entering	Leaving	Objective	PrimalFeasible	DualFeasible	Point
0	primal	1	T	a4	100	yes	no	(0, 0)
1	primal	1			-0	yes	yes	(100, 0)
2	primal	2	e4	s2	700	yes	no	(100, 0)
3	primal	2	C	s1	3,500	yes	no	(500, 0)
4	primal	2			4,040	yes	yes	(320, 360)

Path of the primal simplex



Primal simplex - phase 1, tableau 0
enter T, leave a4 (min ratio 100)

	T	C	s1	s2	s3	e4	a4	RHS
w	-1	0	0	0	0	1	0	-100
s1	3	4	1	0	0	0	0	2400
s2	2	1	0	1	0	0	0	1000
s3	0	1	0	0	1	0	0	450
a4	1	0	0	0	0	-1	1	100

Primal simplex - phase 1, tableau 1
end of phase 1

	T	C	s1	s2	s3	e4	a4	RHS
w	0	0	0	0	0	0	1	0
s1	0	4	1	0	0	3	-3	2100
s2	0	1	0	1	0	2	-2	800
s3	0	1	0	0	1	0	0	450
T	1	0	0	0	0	-1	1	100

Primal simplex - phase 2, tableau 2
enter e4, leave s2 (min ratio 400)

	T	C	s1	s2	s3	e4	RHS
z	0	-5	0	0	0	-7	700
s1	0	4	1	0	0	3	2100
s2	0	1	0	1	0	2	800
s3	0	1	0	0	1	0	450
T	1	0	0	0	0	-1	100

Primal simplex - phase 2, tableau 3
enter C, leave s1 (min ratio 360)

	T	C	s1	s2	s3	e4	RHS
z	0	-1.5	0	3.5	0	0	3500
s1	0	2.5	1	-1.5	0	0	900
e4	0	0.5	0	0.5	0	1	400
s3	0	1	0	0	1	0	450
T	1	0.5	0	0.5	0	0	500

Primal simplex - phase 2, tableau 4
optimal tableau

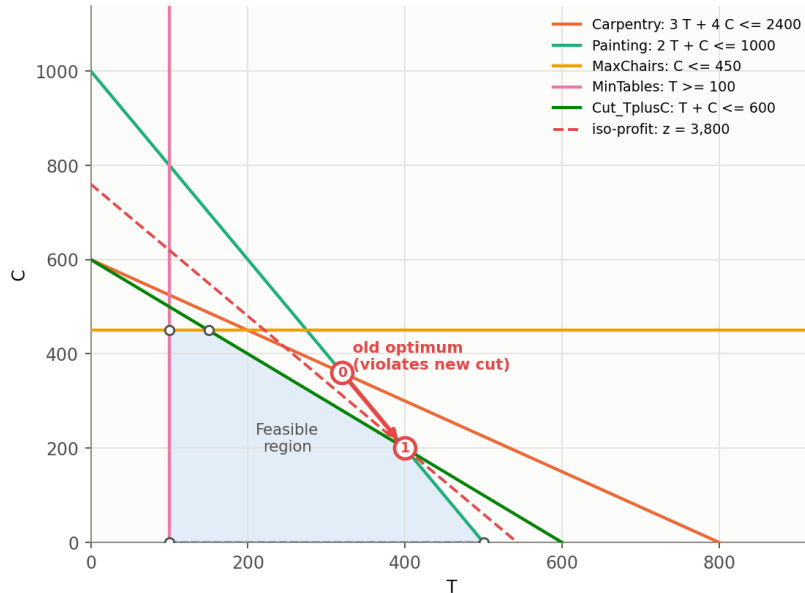
	T	C	s1	s2	s3	e4	RHS
z	0	0	0.6	2.6	0	0	4040
C	0	1	0.4	-0.6	0	0	360
e4	0	0	-0.2	0.8	0	1	220
s3	0	0	-0.4	0.6	1	0	90
T	1	0	-0.2	0.8	0	0	320

Dual simplex after adding 'Cut_TplusC'

New constraint: $1 T + 1 C \leq 600$. dual simplex demo: combined capacity $T + C \leq 600$ Result: **optimal**, new optimum (400, 200), Profit (\$) = 3,800.

Tableau	Method	Phase	Entering	Leaving	Objective	PrimalFeasible	DualFeasible	Point
0	dual	2	s1	s_Cut_TplusC	4,040	no	yes	(320, 360)
1	dual	2			3,800	yes	yes	(400, 200)

Dual simplex: start outside the new region (primal infeasible), end at its optimum



Dual simplex - phase 2, tableau 0
leave s_Cut_TplusC (RHS -80 < 0), enter s1 (min |ratio| 3)

	T	C	s1	s2	s3	e4	s_Cut_TplusC	RHS
z	0	0	0.6	2.6	0	0	0	4040
C	0	1	0.4	-0.6	0	0	0	360
e4	0	0	-0.2	0.8	0	1	0	220
s3	0	0	-0.4	0.6	1	0	0	90
T	1	0	-0.2	0.8	0	0	0	320
s_Cut_TplusC	0	0	-0.2	-0.2	0	0	1	-80

Dual simplex - phase 2, tableau 1
optimal (primal feasible again)

	T	C	s1	s2	s3	e4	s_Cut_TplusC	RHS
z	0	0	0	2	0	0	3	3800
C	0	1	0	-1	0	0	2	200
e4	0	0	0	1	0	1	-1	300
s3	0	0	0	1	1	0	-2	250
T	1	0	0	1	0	0	-1	400
s1	0	0	1	1	0	0	-5	400